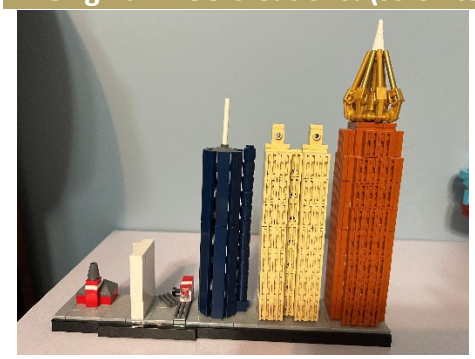
PHILLIP PORTER

MECHANICAL ENGINEER AT GEORGIA INSTITUTE OF TECHNOLOGY

Original LEGO creations! (scroll to next page for engineering stuff)



Atlanta Skyline

- Inspired by LEGO's own **Architecture skyline** sets
- Featuring from left to right: Tech Tower (my school), Flatiron Building (Atlanta), MARTA train (in lieu of a body of water), The Westin, 191 Peachtree Tower, and the Bank of America Plaza.



Avatar the Last Airbender MOCs

- The Avatar the Last Airbender theme retired too soon!
- Featuring from left to right: Avatar Korra, the cabbage cart, Appa, and an Agni Kai.



More Fossils!

- Remember LEGO Ideas 21320 Dinosaur Fossils? I made some more fossils!
- Featuring from left to right: Gigantopithecus, Pachyrhinosaurus, Lystrosaurus, Pachycephalosaurus, Quetzalcoatlus, Compsognathus, Megatherium, Terror Bird, Glyptodon, Dimetrodon, Megaloceros

Extensive LEGO collection



Worldliness

- LEGO is a truly global brand.
- I enjoy building architecture sets for places I've visited.
- When I studied abroad in France last summer, I traveled to **Billund** to visit the **LEGO House**!



Connecting Generations

- My **father** enjoyed playing with LEGO bricks when he was younger. He built a Millenium Falcon before LEGO Star Wars sets existed.
- Now I build towers with my **toddler nephew**. I brought out the bucket of **Duplo** as soon as he was old enough.



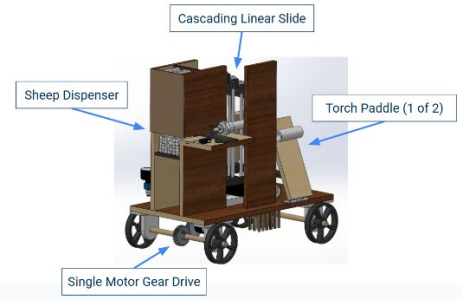
A Passion for STEM

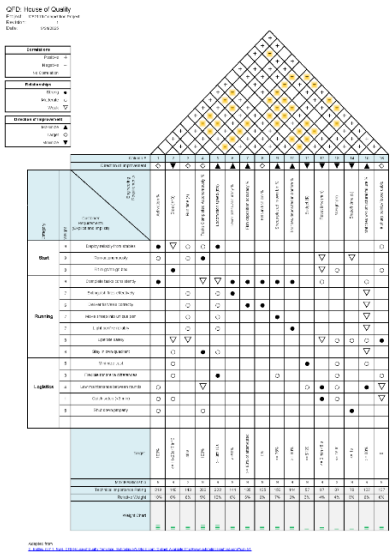
- I attribute my life-spanning interest in **STEM** to my experience with LEGO bricks.
- My first interaction with robotics and coding was a part of a **FIRST LEGO League** team using **MINDSTORMS**. Now I'm a mechanical engineering student with a minor in computer science!

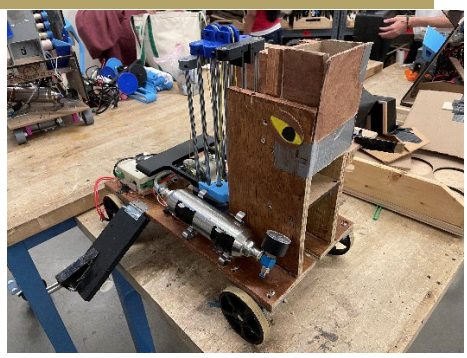
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Competition Robot Team Project: https://youtu.be/GxBOmKxy7oA







What?

- Design and build autonomous robot for Georgia Tech's ME 2110 competition.
- Develop mechanically triggered, **pneumatic** "sheep" dispenser and cascading linear slide elevated-lever actuator subsystems, accounting for **68%** of the robot's scoring potential.

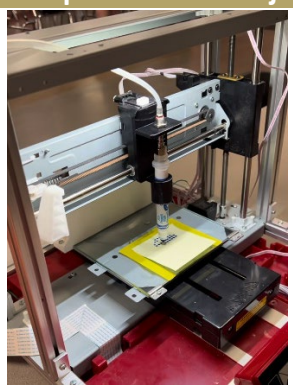
How?

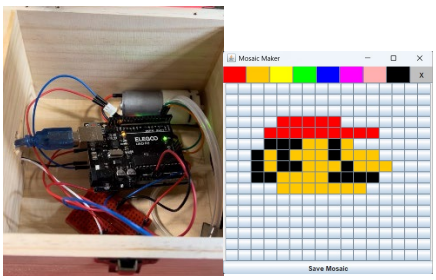
- Created detailed **spec sheets**; performed **quality analysis** and **FMEA**. Programmed robot and designed logic in **Arduino C++.**
- Served as communications lead, coordinated sub teams and ensured consistent documentation and iterative testing over course of semester.

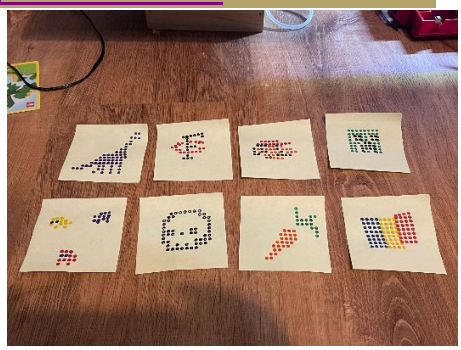
Results

- Achieved 90% accuracy across 20 trials for all subsystems
- Ranked **14th seed** and placed within the top 16 of 70 competing teams.
- Implemented entire engineering design process and teamwork to meet the design challenge.

Pick-and-place Robot Project: https://youtube.com/shorts/bbeM2_WT_ps?feature=share







What?

- Design and assemble a device for demonstrating the utility of gantry systems in assembly manufacturing.
- Pick up markers of different colors and stamp them in custom designs.

How?

- Integrated **pneumatic** vacuum pickup with 3D printer hardware for marker placement.
- Programmed **serial communication** between custom **Java UI**, **Arduino**, and printer for motion and suction control.

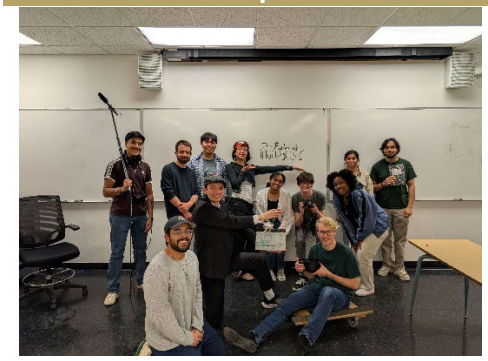
Results

- Presented successful demonstration to a board of judges and received Engineering Star Student award.

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"The Rules of Improv" short film: https://www.youtube.com/watch?v=bnPe7SAxU0A







What?

- Create a theater quality **short film** for production by the Georgia Tech Buzz Studios film club.

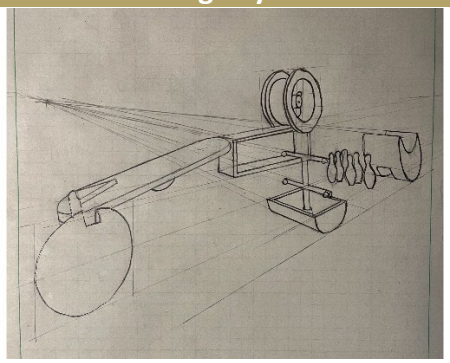
How?

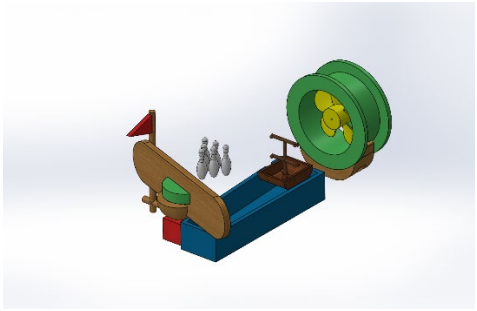
- Wrote, directed, and edited a 9 minute short film, managing a crew of 10 and cast of 10 through full production **pipeline**.
- Coordinated shoot **logistics and deadlines** under Buzz Studio's student production framework.

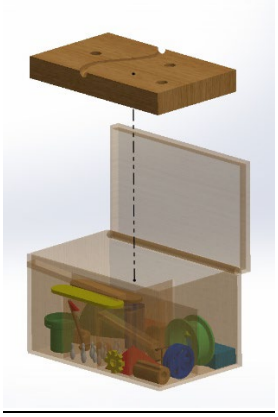
Results

- Premiered the short film to a theater audience on time despite an accelerated schedule of an 8-hour shoot and 1 week of editing.

Rube Goldberg Toy Set: https://youtu.be/9epTtyJC8zA







What?

- Design a STEM educational activity for use in elementary schools.
- Demonstrate advanced **CAD** design and presentation skills.

How?

- **Hand drafted** then **CAD**ded each part of the assembly in **SOLIDWORKS.**
- Performed **motion studies** to ensure parts interact with each other properly in several unique ways.

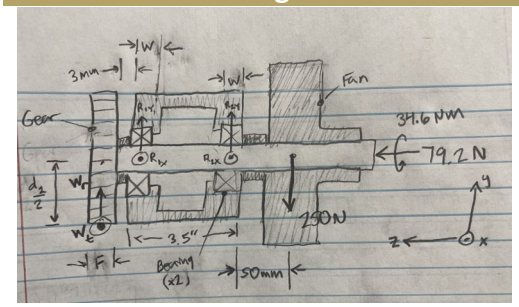
Results

- Created detailed **assembly instructions** for multiple orientations.
- Presented **animated physics simulations** of product in action.

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Fan Machine Design



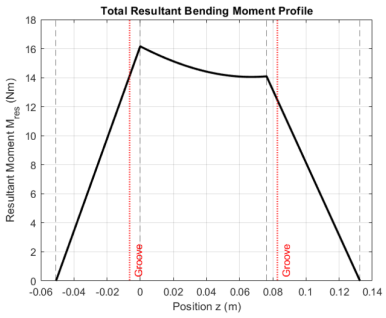


Figure 4: Resultant Moment Diagram.

3.2.2 AGMA Stress Calculations

For the selected face width of $F = 0.606$ in, the stresses are calculated as follows:

Bending Stress (σ) Using the geometry factor $J = 0.376$ (Shigley Figure 14-6, for N=28 meshing with N=60), rim thickness factor $K_b = 1.0$ (Shigley eq. 14-40, for mb≥1.2), and overload factor $K_o = 1.00$ (Shigley figure 14-17 table, Uniform-Uniform ie input torque is essentially constant and load torque varies very little over time):

$$\sigma = \frac{W_t K_o K_v P_d K_m K_b}{FJ} = \frac{(102.08)(1.0)(1.448)(1.00)(10)(1.133)(1.0)}{(0.606)(0.376)} = 7334.59 \text{ psi}$$

Contact Stress (σ_c) Using the elastic coefficient $C_p = 2300\sqrt{\text{psi}}$ (Shigley Table 14-8 for steel pinion steel gear), surface condition factor $C_f = 1.0$, and geometry factor $I = 0.110$:

$$I = \frac{\cos(\phi)\cdot\sin(\phi)}{2}\frac{m_G}{m_G+1} \text{ (Shigley eq. 14-23, for external spur gears)}$$

$$= \frac{\cos(20°)\cdot\sin(20°)}{2}\frac{2.14}{2.14+1} = 0.110$$

$$\sigma_c = C_p\sqrt{\frac{W_t K_o K_v K_s K_m C_f}{d_1 F I}} = 2300\sqrt{\frac{(102.08)(1.0)(1.448)(1.00)(1.133)(1.0)}{(2.8)(0.606)(0.110)}} = 68959.25 \text{ psi}$$

What?

- Propose several fan designs including design of a **support strut**, **rotary shaft**, and **drive gears** as well as **pin**, **retaining ring,** and **machine key** connectors.
- Present rigorous machine design calculations and diagrams and submit **McMaster-Carr** part numbers for a plausible design.

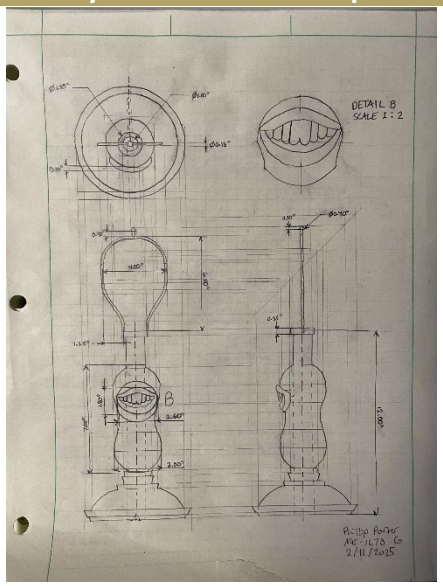
How?

- Designed according to Shigley's Mechanical Engineering Design, 11th Edition.
- Performed all calculations and constructed figures in configurable **MATLAB** scripts.
- Formatted **46 pages** of detailed engineering documents using a **LaTeX** editor.

Results

- Created a feasible design and **BOM** according to the instructor's standards.
- Received an A average across the 3-part project series.

Jimmy Carter Peanut Lamp







What?

- Design a piece of functional home décor that represents my cultural heritage.
- Demonstrate **CAD** techniques.

How?

- Hand drafted each part and the assembly.
- **CAD**ed in **SOLIDWORKS** using all of the basic modeling techniques as well as advanced **surface texture.**

Results

- **3D printed** a half-scale prototype.
- Demonstrated a wide array of **CAD** skills in a single product.